# DeepNeoAG Learning Through Controlled Experiments

Computational Biology
Course investigation

Complete experimental run
25 October 2026

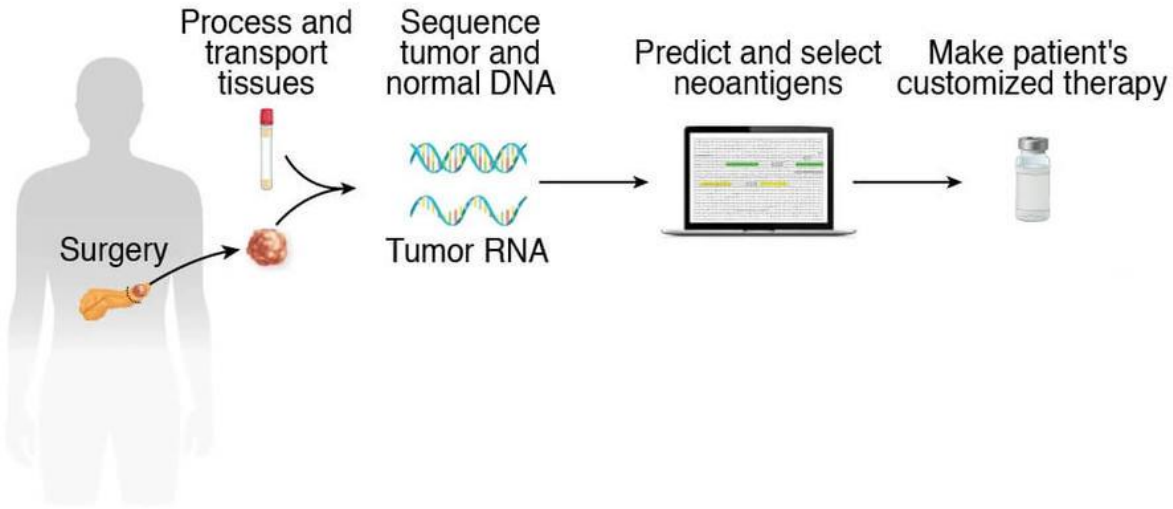
Composition + LR / One-hot MCNN / ESM pooled LR / ESM CNN / ESM MCNN

# Introduction

# Why predict neoantigen-like peptides?

## Sequence prediction covers one part of a wider biological process

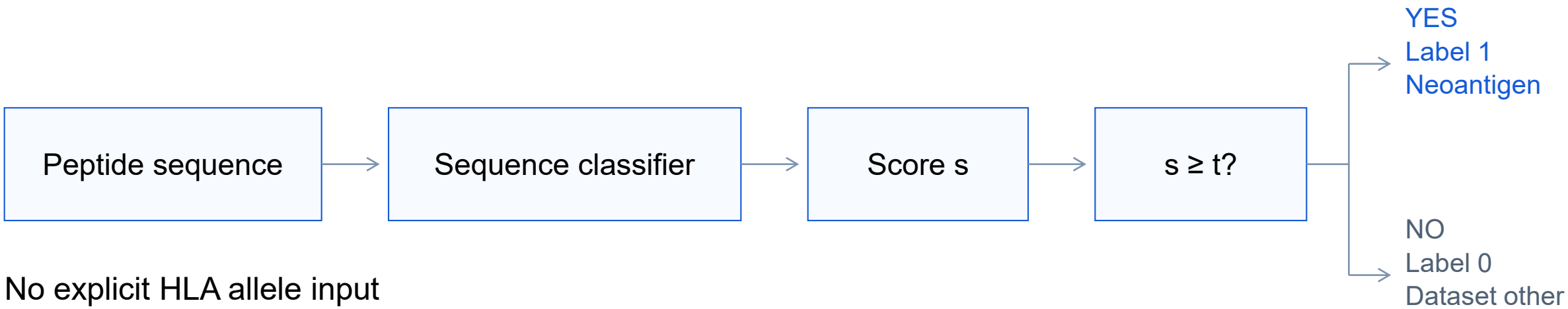
Tumor-specific mutations can alter peptides. Presentation and T-cell recognition depend on biological context.




Biological context only. This classifier does not establish presentation, immune response or vaccine effectiveness.

# Defining the prediction problem

## A ranking score becomes a binary decision at a locked threshold




No explicit HLA allele input

t selected from training OOF predictions

A score is not proof of immunogenicity or a calibrated probability. Label 0 reflects the dataset definition.

# Five questions behind the model sequence

Each approach is trained independently

| Approach | Research question |
| --- | --- |
| Composition + LR | Are residue frequencies sufficient? |
| One-hot + MCNN | What can ordered learning from labels achieve? |
| ESM pooled LR | Do frozen pretrained representations add signal? |
| ESM CNN | Does local extraction help contextual embeddings? |
| ESM MCNN | Do multiple local scales help, with added capacity? |

Composition versus one-hot changes representation and classifier. Later matched comparisons isolate components more closely.

# Methodology

# Dataset provenance and published splits

## The independent test keeps its published boundary

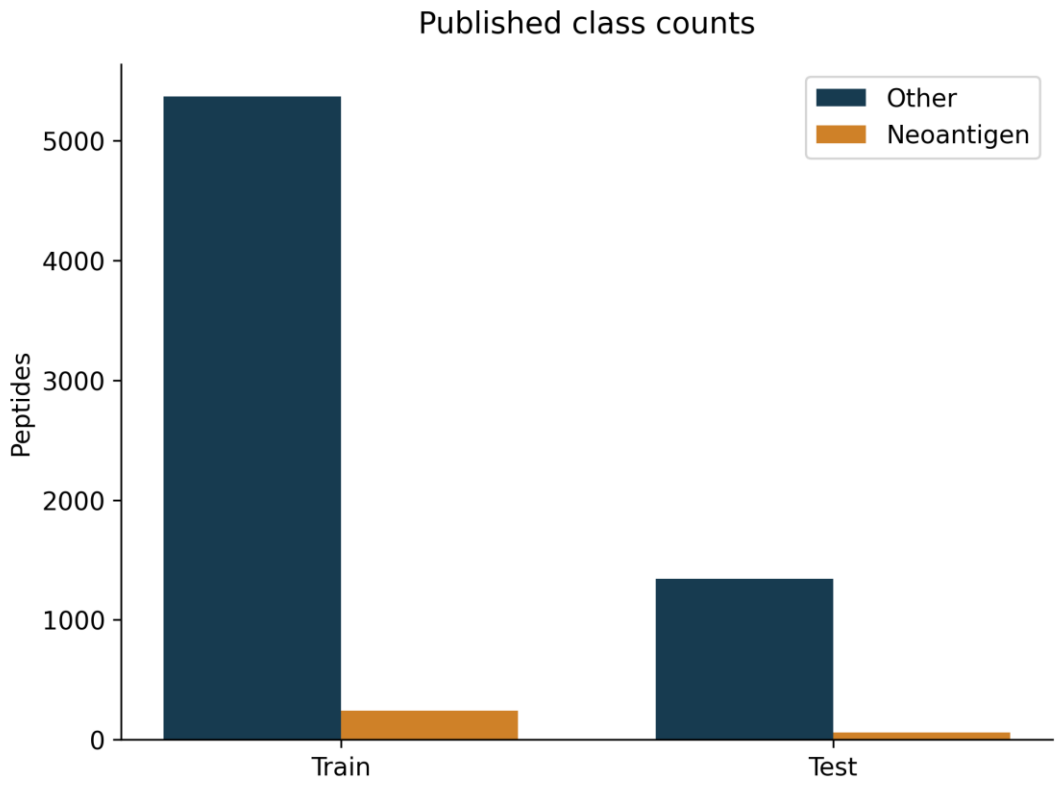


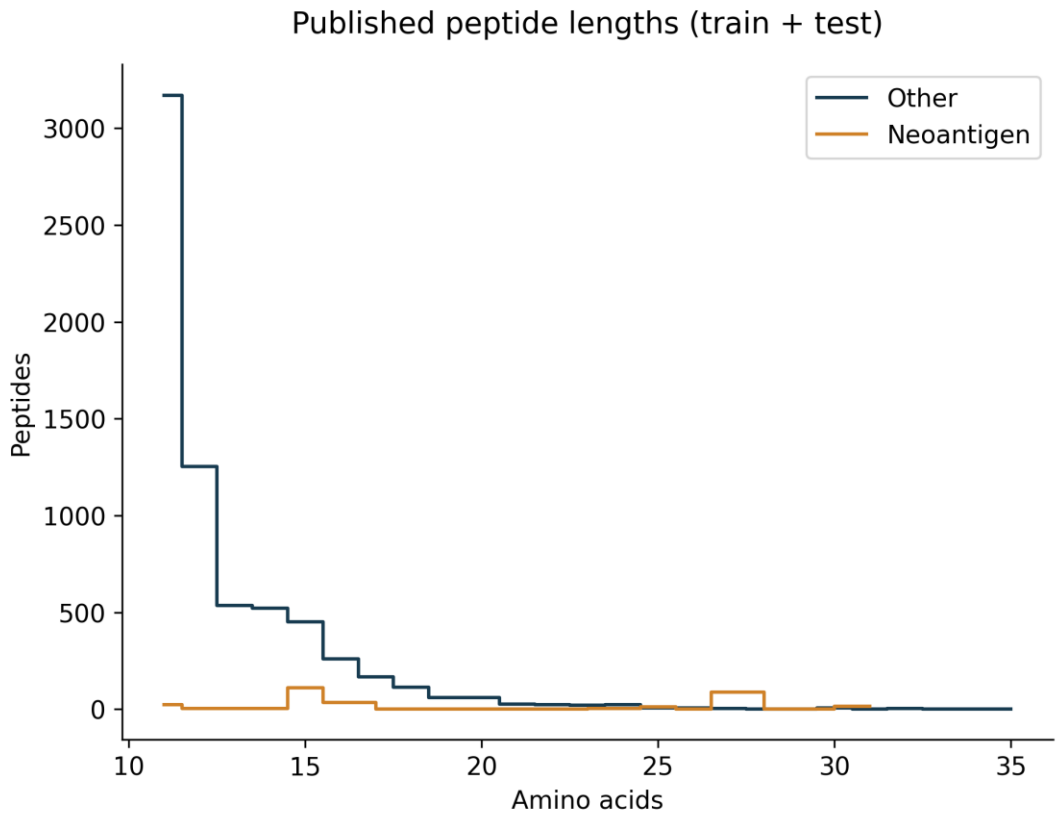

| Partition | Neoantigen (1) | Other (0) | Total |
|---|---|---|---|
| Training | 241 | 5,367 | **5,608** |
| Independent test | **61** | 1,343 | **1,404** |
| Total | 302 | 6,710 | 7,012 |

Record IDs and source labels are retained. Source similarity filtering has not been independently reconstructed.

# Understanding the data before training

## Rare positives and a length-35 boundary case affect the method

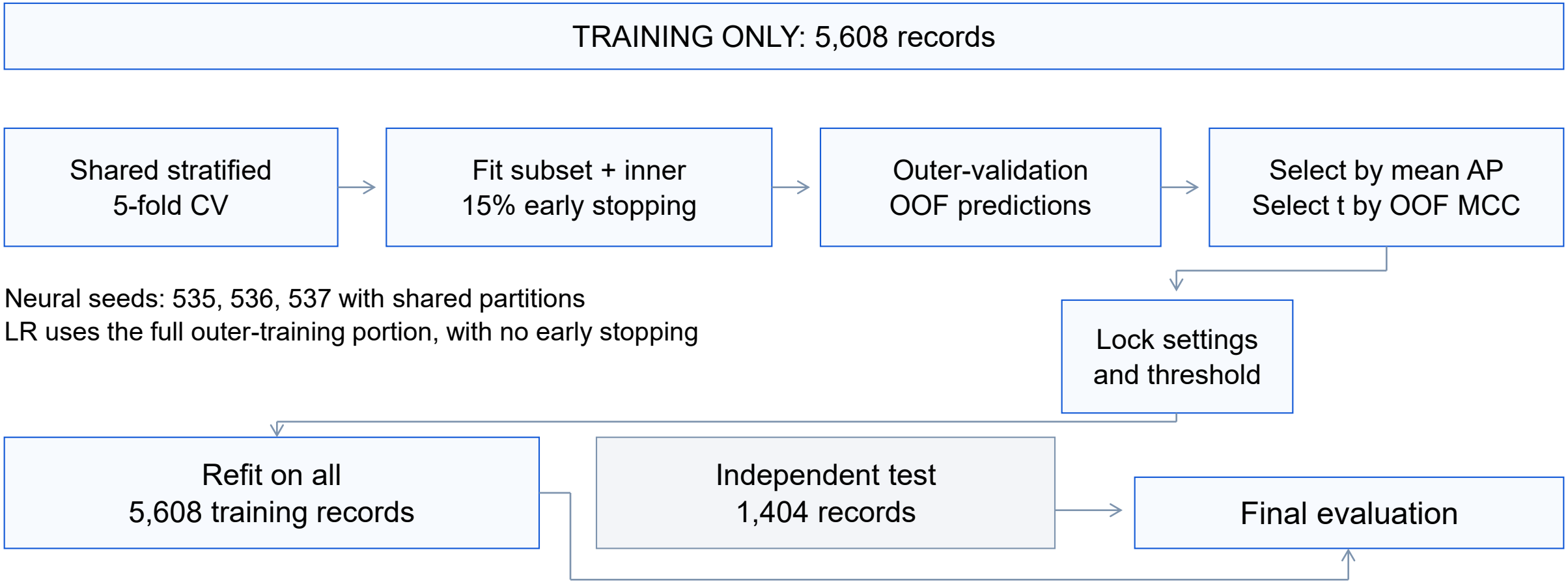



About 1 positive per 22 others. Keep the length-35 record. Length alone does not establish an MHC class.

# Designing a fair evaluation

## Fitting, early stopping, model selection and testing use distinct roles




Selection CV is optimistic after tuning. Seeds on the same fold are correlated evaluations, not independent datasets.

# Metric definitions and threshold policy

## ranking quality and a binary operating point answer different questions

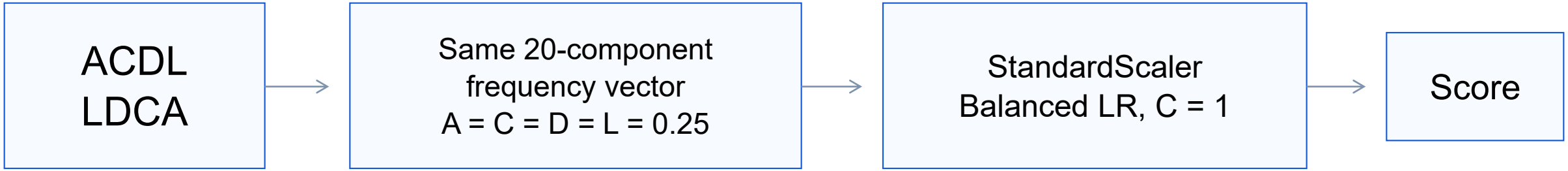
| Measure | Meaning |
|---|---|
| Average precision (AP) | Summarizes precision across recall changes in a ranked list |
| ROC-AUC | Ranking discrimination across false-positive-rate levels |
| Precision | $TP / (TP + FP)$ |
| Recall / sensitivity | $TP / (TP + FN)$ |
| Specificity | $TN / (TN + FP)$ |
| MCC | $(TP \times TN - FP \times FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$ |

Configuration: maximize mean CV AP. Threshold: maximize MCC from seed-averaged training OOF predictions. Independent test uses both 0.5 and the locked OOF threshold. Main slides use the OOF threshold.

AP and ROC-AUC use scores. Precision, recall, specificity, MCC and confusion counts depend on the threshold.

# Baseline 1: amino-acid composition

## Composition contains some signal, but loses residue order




## Order is lost

| Mean CV AP | Test AP | Test MCC | OOF threshold |
|---|---|---|---|
| 0.1287 | **0.0714** | 0.0515 | 0.6508 |

ACDL and LDCA are teaching examples. Test prevalence: 0.0434. The scaler fits within training partitions.

# Baseline 2: one-hot MCNN

## Ordered local patterns learned directly from task labels

No pretrained encoder

Five parallel windows
k = 2, 3, 4, 5, 8

64 filters per branch
320 concatenated features

Mask padding before max pooling




| Trainable parameters | CV AP | Test AP | Test MCC |
|---|---|---|---|
| **28,801** | 0.4943 | **0.5408** | 0.5495 |

Strong efficiency reference. Improvement over composition changes both representation and classifier.

# From peptides to contextual embeddings

## A frozen ESM encoder gives every real residue a contextual representation

IMAGE PLACEHOLDER
One-hot, embeddings and tensor layout
Prompt C in IMAGE_GENERATION_PROMPTS.md

B
Peptides per batch

L
Padded residue positions

D
Features per residue

8M: D = 320
35M: D = 480

Remove special tokens. Mean pooling uses real residues. CNN windows touching padding are excluded.

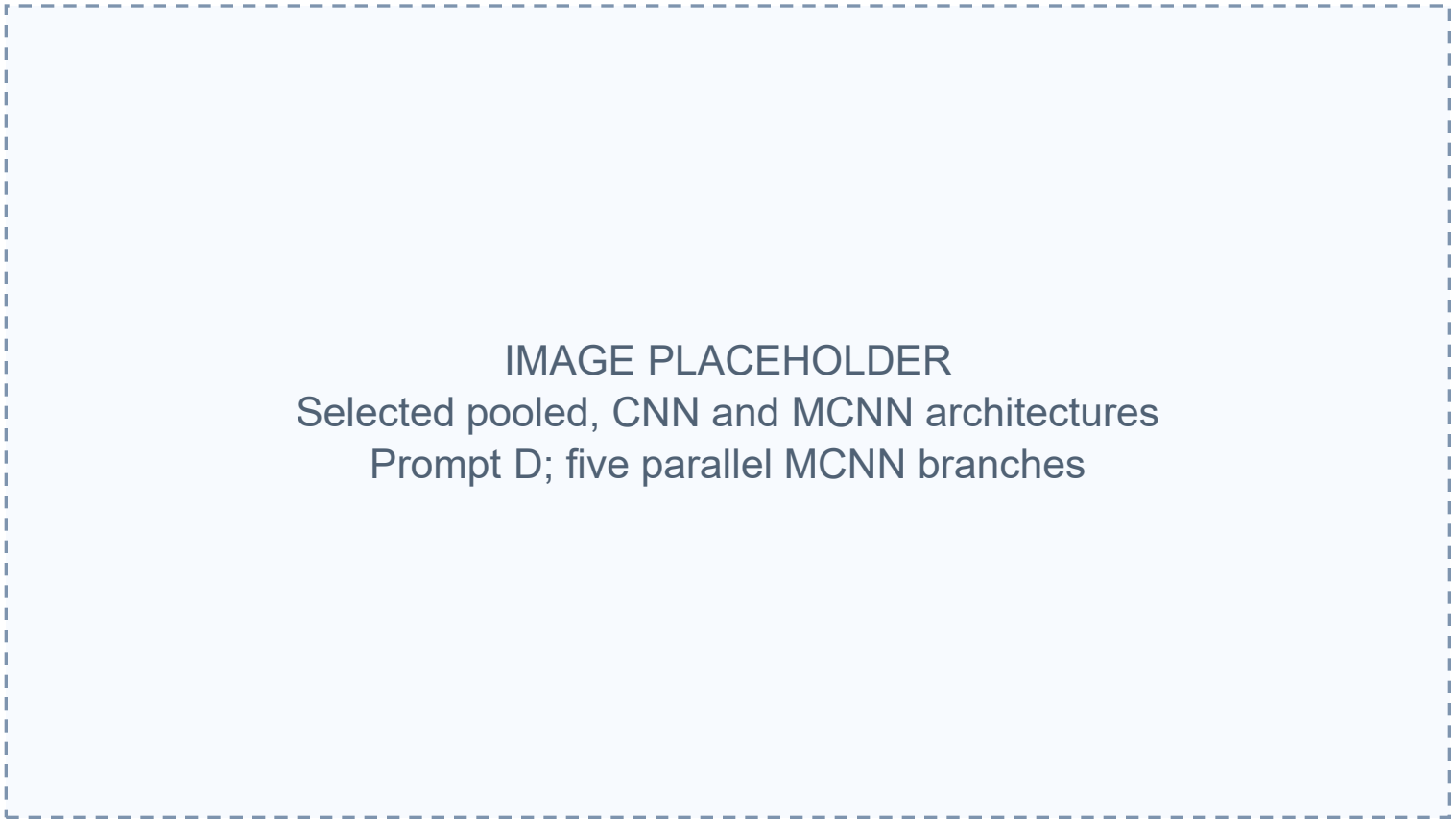
# Pooling, single-window CNN and MCNN

## The same frozen embeddings support three independent classifiers

Pooled LR
Real-residue mean; C = 0.1
CV / test AP: 0.3976 / 0.4764

Single-window CNN
k = 3; 64 filters
CV / test AP: 0.4868 / 0.5122

Multi-window CNN
256 filters × 5; concat 1,280
CV / test AP: 0.5142 / 0.5437




MCNN adds capacity as well as window diversity. Contextual mean pooling retains information learned by ESM.

# Our controlled experiment matrix

## A staged, one-factor search, with 22 configurations

| Changed component | Values | Held fixed / caveat |
|---|---|---|
| LR regularization | C = 0.1, 1, 10 | Balanced LR; shared folds |
| Single CNN window, 8M | k = 2, 3, 4, 5, 8 | 64 filters; weighted loss |
| MCNN filters, 8M | 32, 64, 128, 256 per branch | Same five windows; weighted loss |
| MCNN loss, 8M | Weighted / unweighted | 64 filters; same windows |
| Frozen encoder | 8M / 35M | Transfer selected CNN settings; D and head size change |

240 CV fits + 18 final refits = 258 fits. AdamW, LR 0.001, weight decay 0.01, batch 64, maximum 30 epochs.

# Selected configurations

## settings locked before independent testing

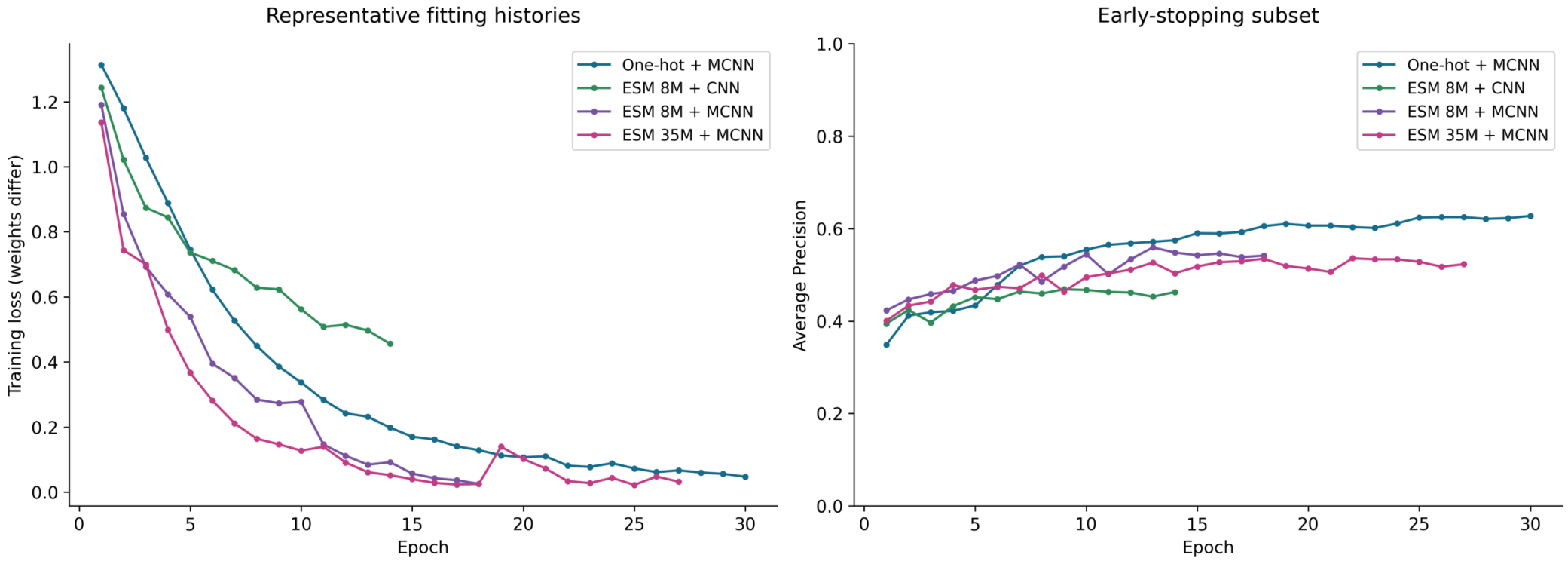
| Model | Selected setting | Parameters | OOF t | Refit epochs |
|---|---|---|---|---|
| Composition + LR | C = 1 | 21 | 0.650830 | Not applicable |
| One-hot + MCNN | k=2,3,4,5,8; 64 filters | **28,801** | 0.512465 | 23 |
| ESM 8M + pooled LR | C = 0.1 | 321 | 0.977457 | Not applicable |
| ESM 8M + CNN | k=3; 64 filters | 61,569 | 0.930151 | 13 |
| ESM 8M + MCNN | **k=2,3,4,5,8; 256 filters** | 1,804,801 | 0.766193 | 16 |
| ESM 35M + pooled LR | C = 0.1 | 481 | 0.945360 | Not applicable |
| ESM 35M + CNN | k=3; 64 filters | 92,289 | 0.732428 | 10 |
| ESM 35M + MCNN | **k=2,3,4,5,8; 256 filters** | **2,705,921** | 0.632288 | 14 |

ESM is frozen. Parameters count only the trainable classifier. Neural final predictions average seeds 535, 536, 537.

# Results

# Tracking learning and computational cost

## Early-stop AP can plateau while fitting loss continues to decline




**Extraction (8M / 35M)**

9.53 s / 13.21 s

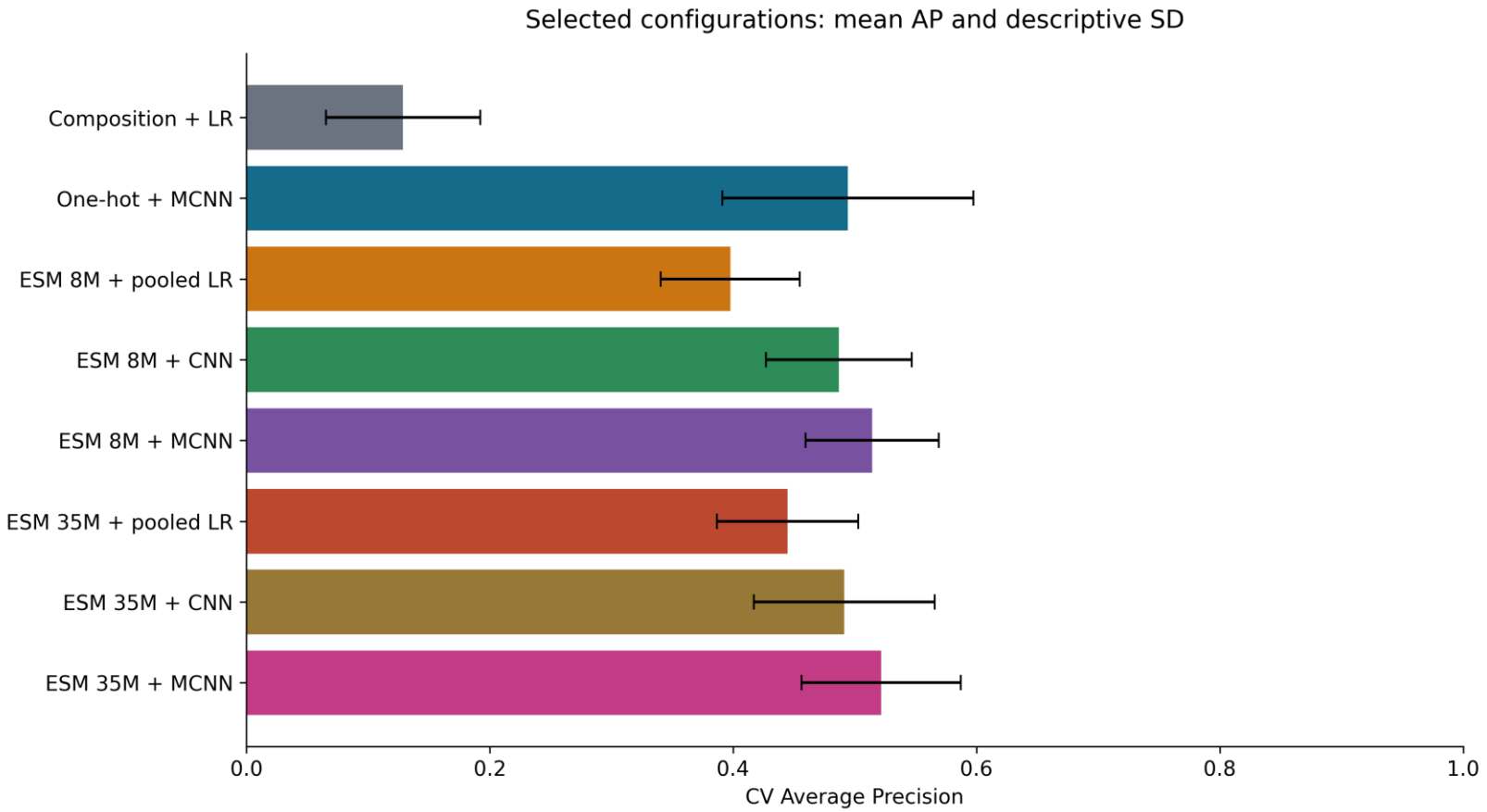
**Mean MCNN CV fit (one-hot / 8M / 35M)**

10.26 s / 11.50 s / 13.52 s

Fold 0, seed 535. Raw loss scales differ. Timings depend on session state and stopping epoch, not deployment latency.

# What we observed across folds and seeds

## Mean AP and descriptive spread answer different questions




### Mean CV AP ± SD

One-hot MCNN
0.4943 ± 0.1032

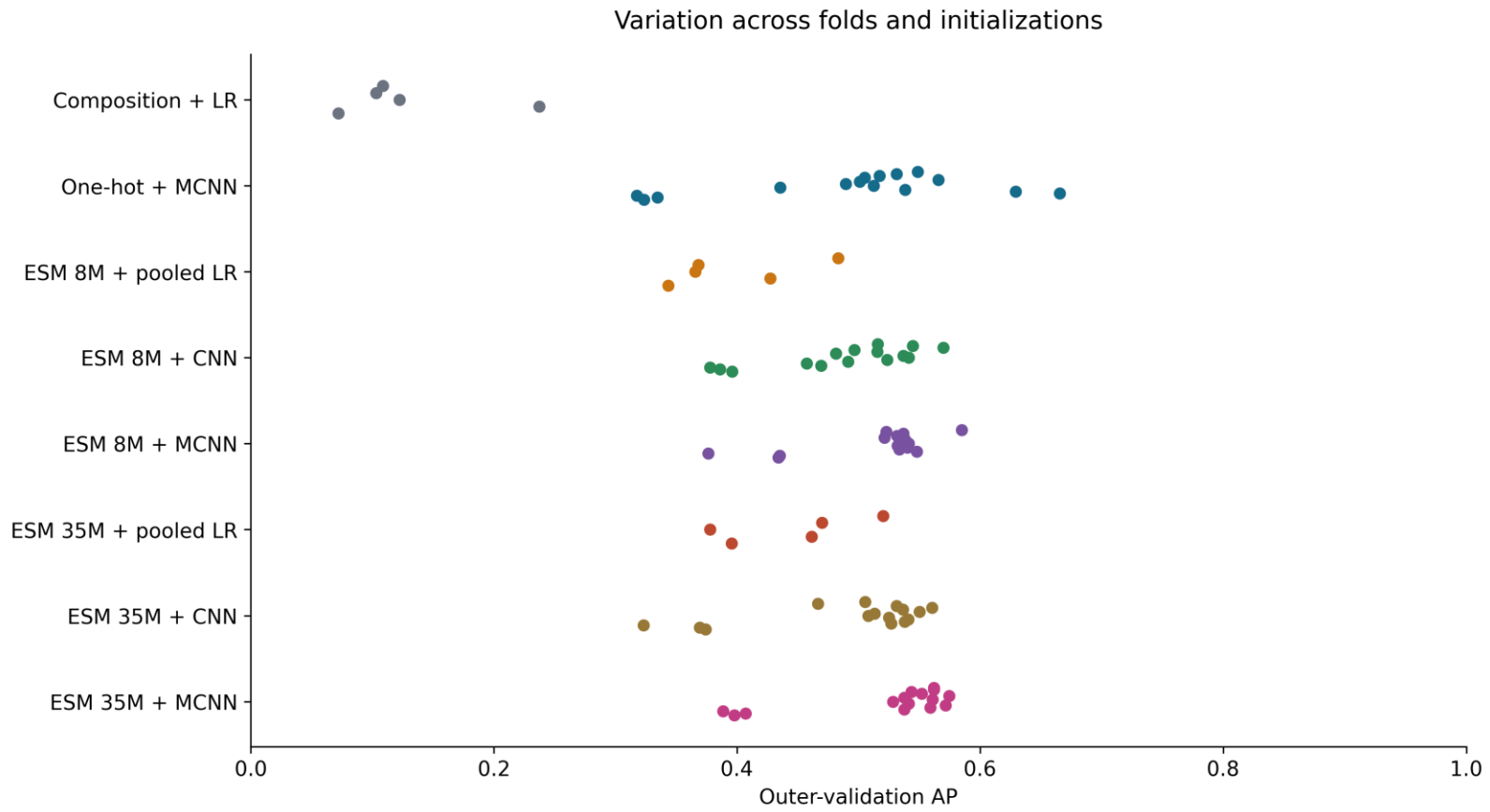
8M MCNN
0.5142 ± 0.0547

35M MCNN
0.5216 ± 0.0655

Selection CV, not an unbiased post-selection estimate. Fold 4 is difficult across neural methods; the cause is unresolved.

# Fold and seed variation

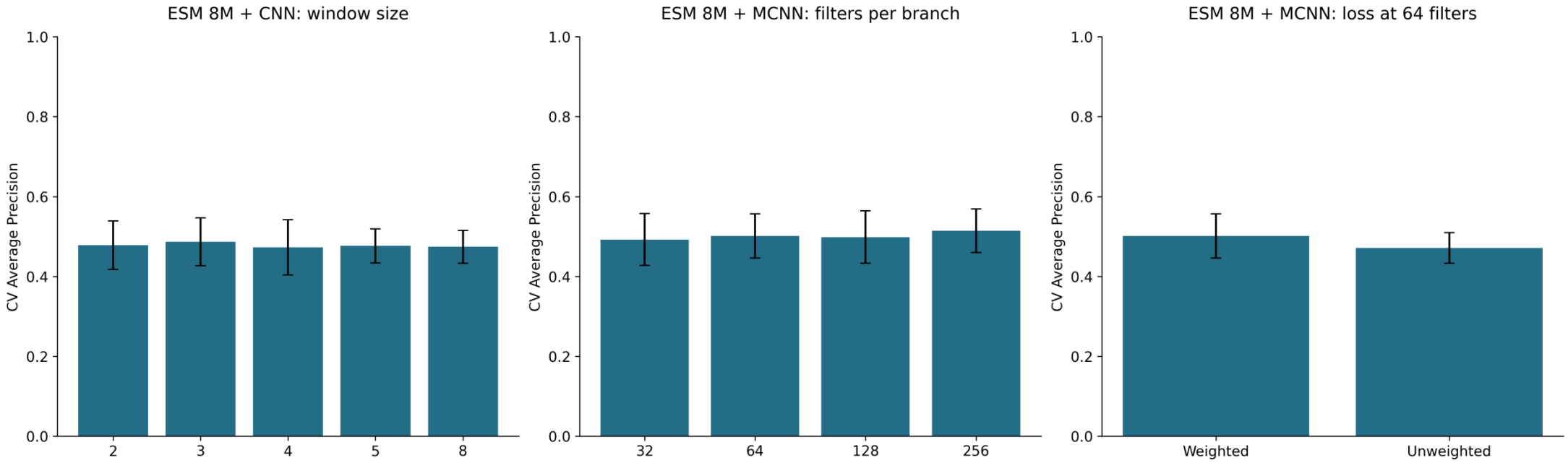
each point is one run on the same published training split




Seeds and folds are correlated. Spread is descriptive, not a confidence interval. Difficult fold causes remain unresolved.

# What changed when one component changed?

## Ablations describe conditional effects under the same protocol




**Window**

**k = 3 selected**
**Best–worst mean AP: 0.0138**

**Capacity**

**AP does not rise monotonically**
**32 / 64 / 128 / 256 filters**

**Loss at 64 filters**

**Weighted AP: 0.5013**
**Unweighted AP: 0.4713**

Error bars show descriptive fold/seed SD. Weighted loss improves AP here, while MCC at 0.5 decreases (0.4225 vs 0.4398).

# Fixed defaults versus tuned selections

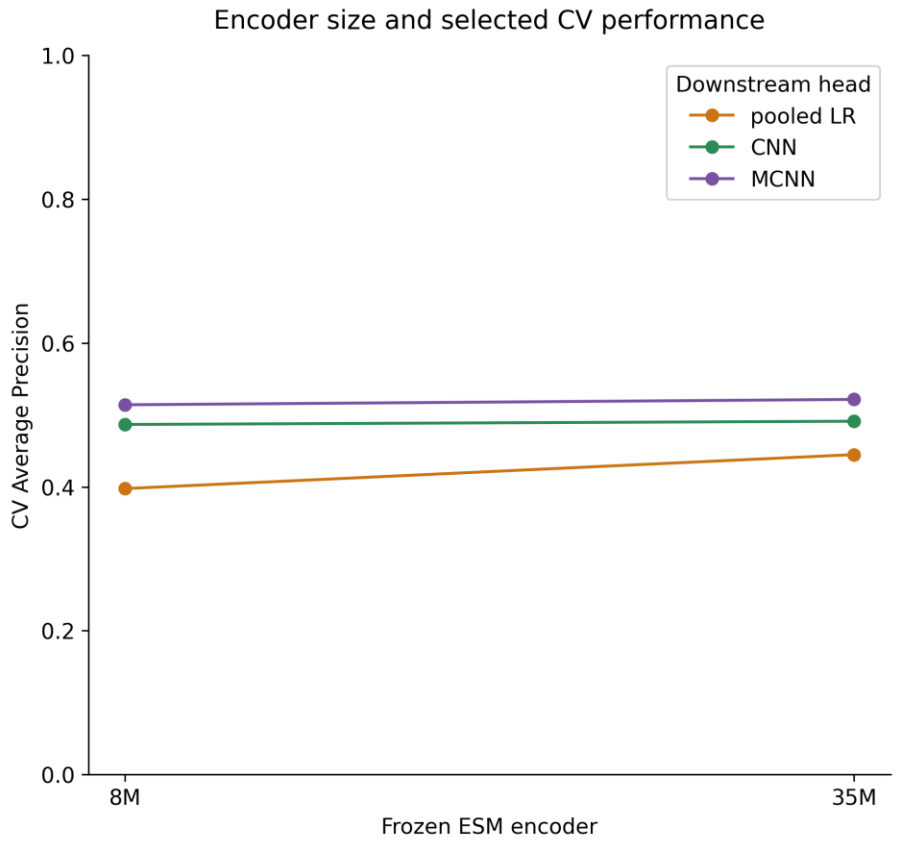
## comparison includes the model-selection step

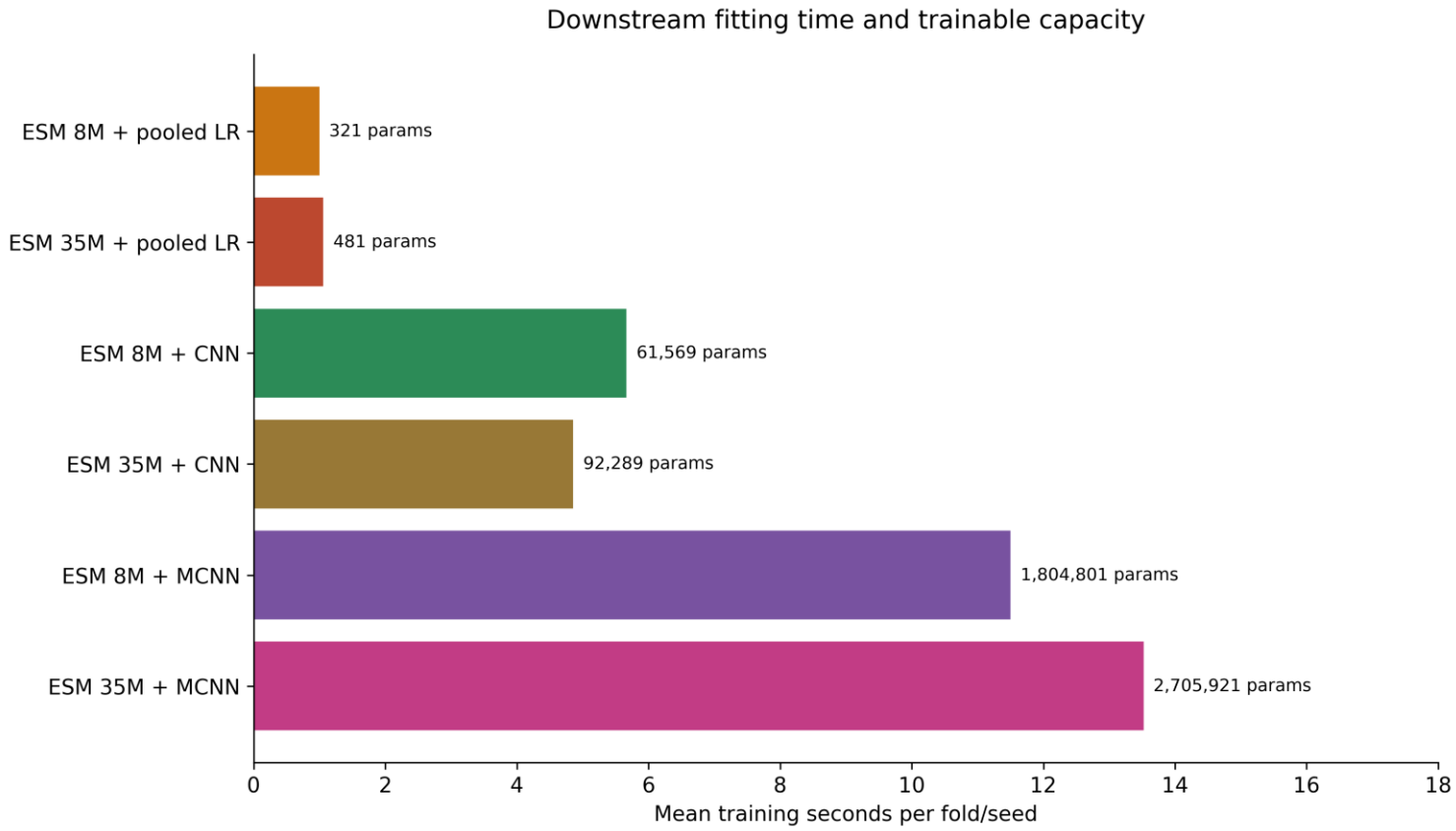
| Model | Default CV AP | Selected CV AP | Default params | Selected params |
|---|---|---|---|---|
| Composition + LR | 0.1287 | 0.1287 | 21 | 21 |
| One-hot + MCNN | 0.4943 | 0.4943 | 28,801 | 28,801 |
| ESM 8M + pooled LR | 0.3264 | **0.3976** | 321 | 321 |
| ESM 8M + CNN | 0.4868 | 0.4868 | 61,569 | 61,569 |
| ESM 8M + MCNN | 0.5013 | **0.5142** | 451,201 | **1,804,801** |

Defaults and selected settings are not independent estimates. A larger selected head changes capacity as well as window coverage.

# Encoder size, head quality and cost

## transferred CNN settings, with changed D and head parameter count

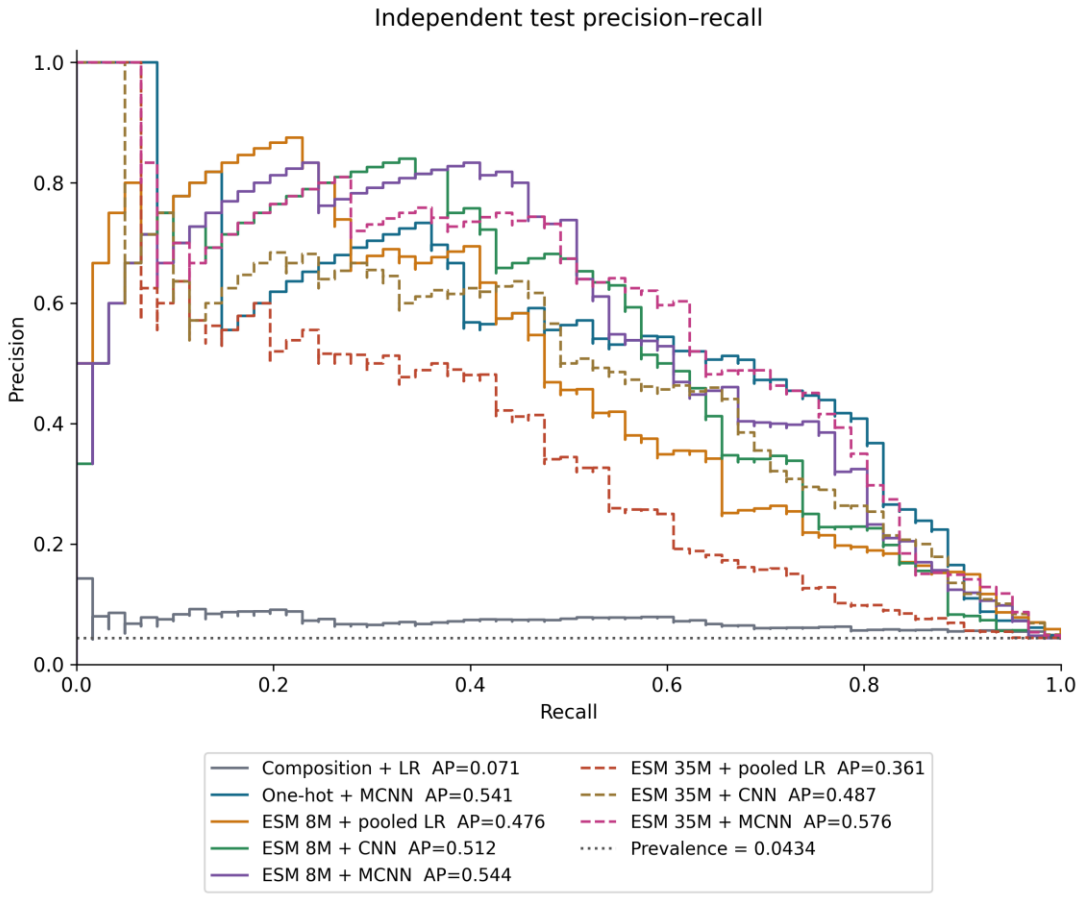



8M / 35M change embedding dimension and head capacity. Downstream fit time excludes extraction and depends on hardware.

# Independent test: a final check

## ESM 35M MCNN has the highest observed AP, with uncertain differences




| Locked model | AP | MCC |
|---|---|---|
| One-hot MCNN | 0.5408 | 0.5495 |
| 8M MCNN | 0.5437 | 0.5512 |
| 35M MCNN | **0.5758** | **0.5513** |

35M MCNN: 44 / 61 positives found
52 false positives

AP 95% CI: [0.4631, 0.7061]

Paired AP difference vs one-hot
95% CI: [−0.0444, 0.1151]

All 1,404 test records. Neural probabilities average three final seeds. Thresholds are locked from OOF MCC.

# Complete independent-test metrics

## ensemble probabilities and locked OOF thresholds

| Model | AP | ROC-AUC | MCC | Precision | Recall | Specificity | TP / FP |
|---|---|---|---|---|---|---|---|
| Composition + LR | 0.0714 | 0.6625 | 0.0515 | 0.0690 | 0.2295 | 0.8593 | 14 / 189 |
| One-hot + MCNN | 0.5408 | 0.9179 | 0.5495 | 0.5362 | 0.6066 | 0.9762 | 37 / 32 |
| ESM 8M + pooled LR | 0.4764 | 0.9077 | 0.4743 | 0.6667 | 0.3607 | 0.9918 | 22 / 11 |
| ESM 8M + CNN | 0.5122 | 0.8880 | 0.5262 | **0.7143** | 0.4098 | **0.9926** | 25 / 10 |
| ESM 8M + MCNN | 0.5437 | 0.9108 | 0.5512 | 0.6000 | 0.5410 | 0.9836 | 33 / 22 |
| ESM 35M + pooled LR | 0.3614 | 0.8213 | 0.3693 | 0.5143 | 0.2951 | 0.9873 | 18 / 17 |
| ESM 35M + CNN | 0.4870 | 0.9154 | 0.4938 | 0.4557 | 0.5902 | 0.9680 | 36 / 43 |
| ESM 35M + MCNN | **0.5758** | **0.9224** | **0.5513** | 0.4583 | **0.7213** | 0.9613 | **44 / 52** |

N = 1,404, including 61 positives and 1,343 others. LR has one refit. Neural probabilities average three refits.

# Test uncertainty and seed spread

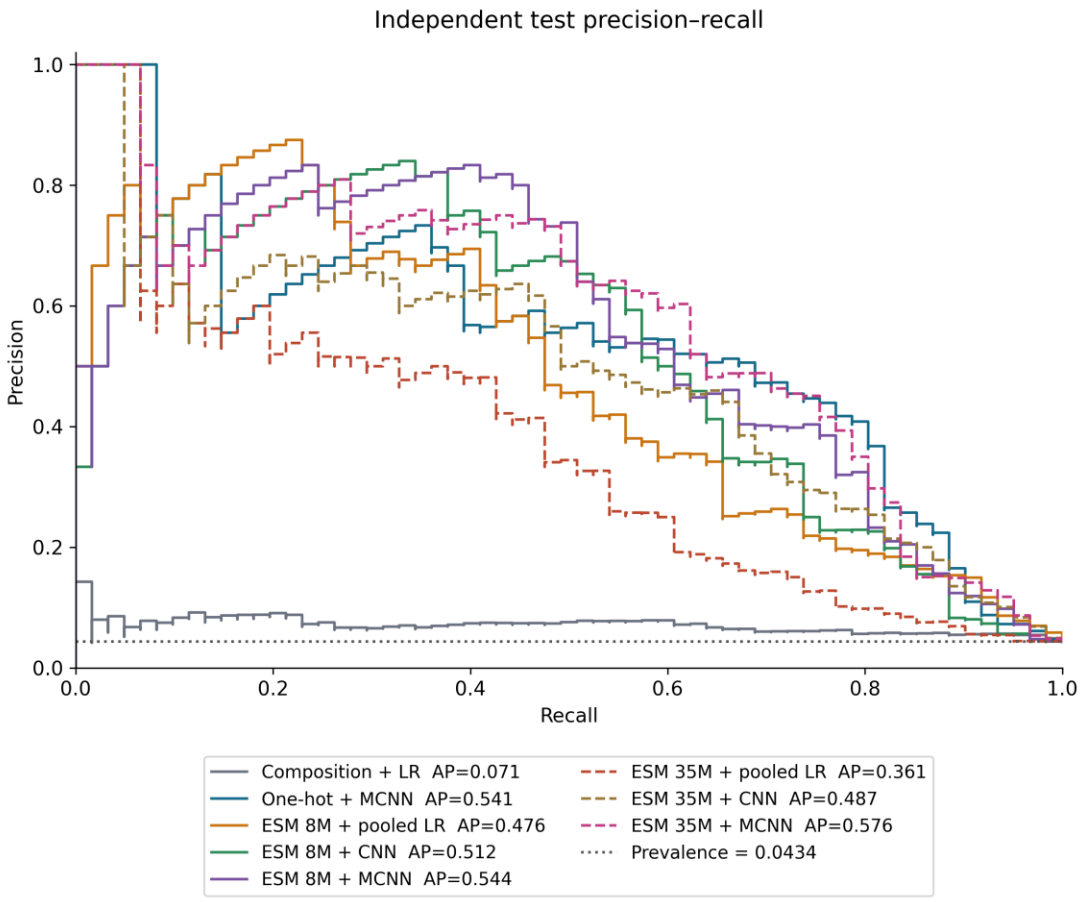
## 2,000 stratified bootstrap repetitions, fixed trained models

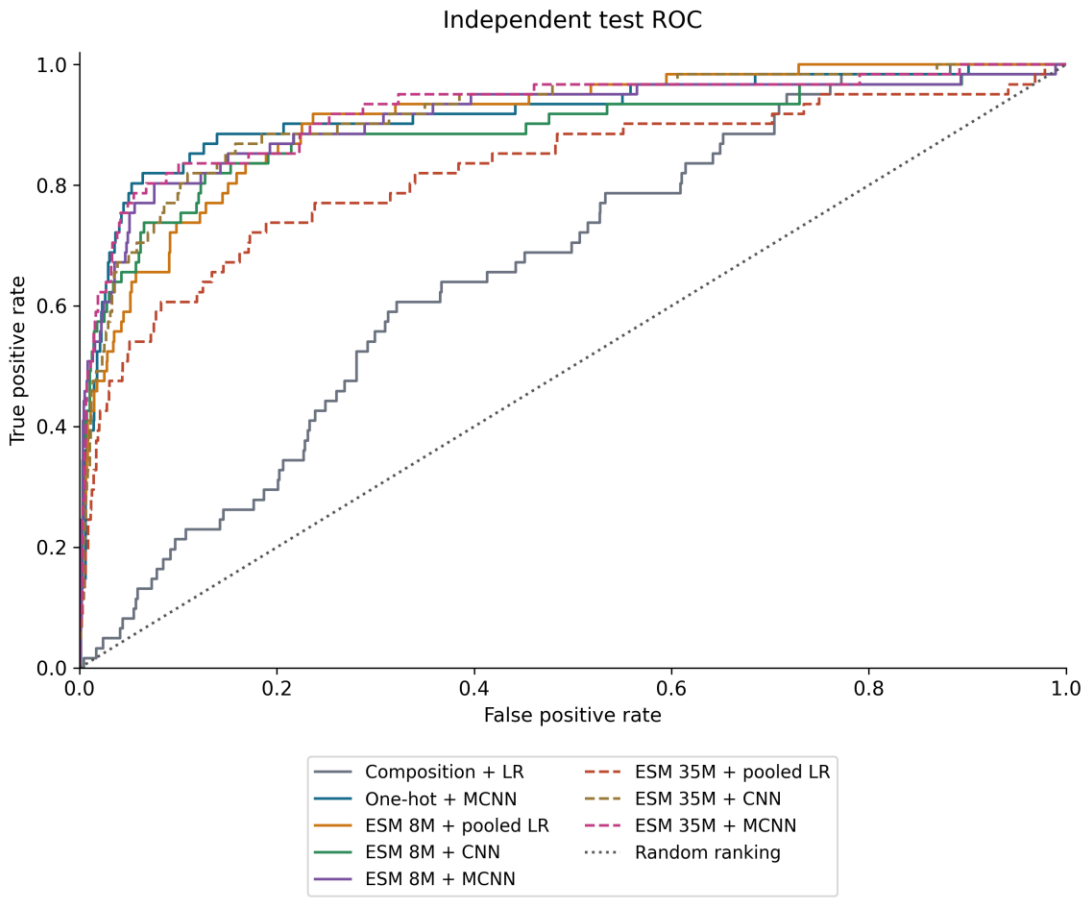
| Model | Test AP | AP 95% interval | MCC 95% interval | Single-seed AP range |
|---|---|---|---|---|
| Composition + LR | 0.0714 | [0.0577, 0.1057] | [-0.0052, 0.1140] | 0.0714–0.0714 |
| One-hot + MCNN | 0.5408 | [0.4347, 0.6662] | [0.4465, 0.6465] | 0.5112–0.5373 |
| ESM 8M + pooled LR | 0.4764 | [0.3655, 0.6187] | [0.3517, 0.5934] | 0.4764–0.4764 |
| ESM 8M + CNN | 0.5122 | [0.4025, 0.6606] | [0.4026, 0.6394] | 0.5094–0.5100 |
| ESM 8M + MCNN | 0.5437 | [0.4322, 0.6933] | [0.4479, 0.6544] | 0.5211–0.5328 |
| ESM 35M + pooled LR | 0.3614 | [0.2573, 0.4934] | [0.2484, 0.4912] | 0.3614–0.3614 |
| ESM 35M + CNN | 0.4870 | [0.3786, 0.6198] | [0.3998, 0.5905] | 0.4667–0.4966 |
| ESM 35M + MCNN | **0.5758** | **[0.4631, 0.7061]** | [0.4681, 0.6338] | 0.5696–0.5833 |

Exploratory paired AP difference, 35M MCNN minus one-hot MCNN: 0.0350, 95% CI [−0.0444, 0.1151].
No multiplicity correction. These intervals do not cover retraining or source-domain uncertainty.

# Complete test precision–recall and ROC

## all eight locked methods on the same independent test

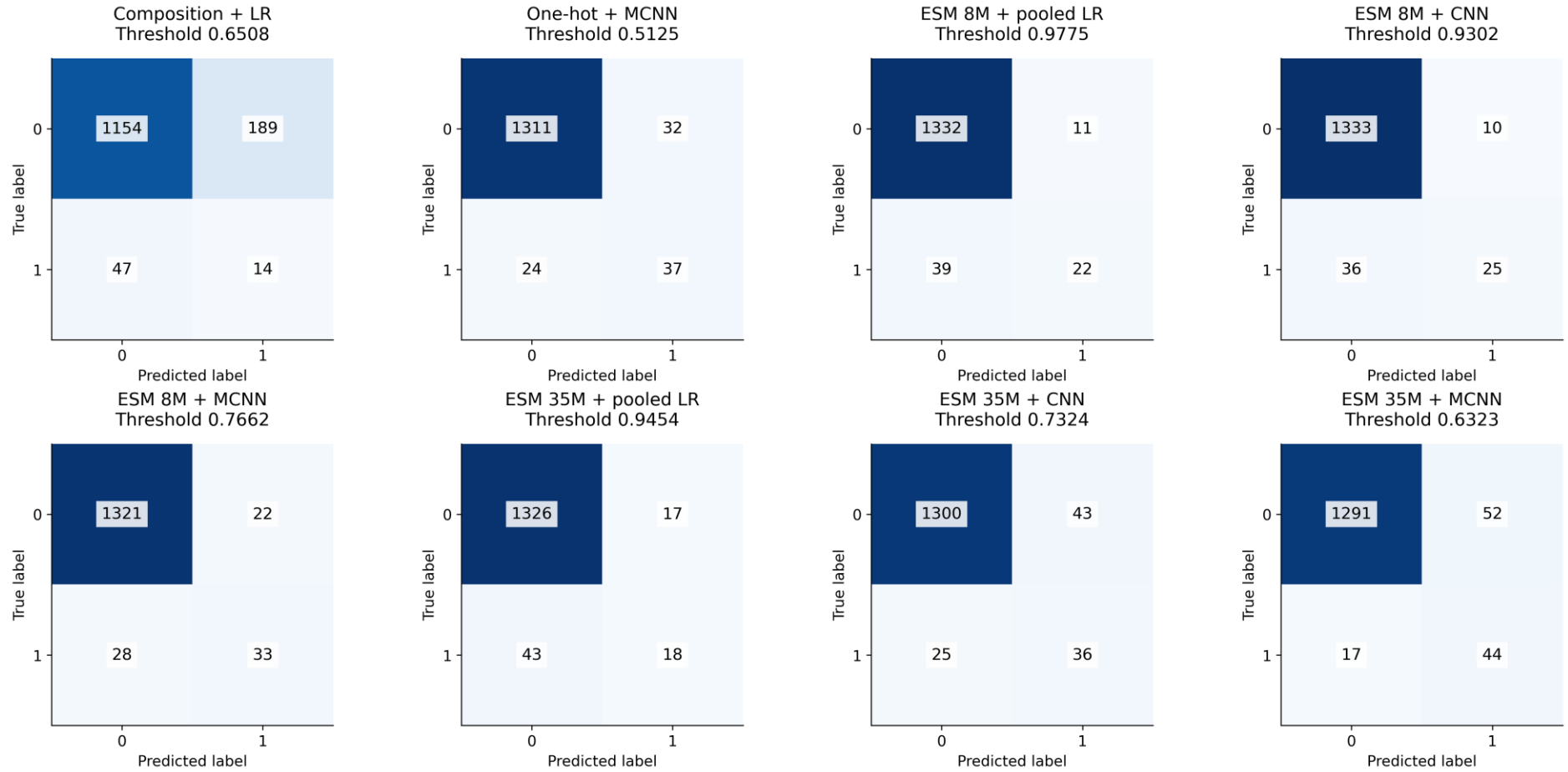



N = 1,404. PR prevalence = 0.0434. Neural scores average three seed refits. No test-driven configuration changes.

# All eight confusion matrices

## raw counts at locked OOF-selected thresholds




Label 1: neoantigen. Label 0: dataset other. Common color scale. Every panel covers the same 1,404 records.

# Conclusion

# Errors, limitations and lessons learned

## Confident errors require inspection without inventing a biological mechanism

| Record ID | Length | Label | Score | Error |
|---|---|---|---|---|
| sgh_sequence7831 | **35** | 0 | **0.999930** | FP |
| neo_sequence56 | 11 | 1 | **0.000162** | FN |

ESM 35M + MCNN
Locked threshold: 0.632288

Only 61 positive test records

Source similarity filtering remains unverified

Labels and source domain limit generalization

Sequence appearance does not identify an error mechanism

Examples selected reproducibly by confidence within FP/FN groups. A length-35 error alone does not prove a padding defect.

# Execution provenance and limitations

numerical reconciliation does not establish exact execution-source identity

258 completed fits and saved numerical evidence reconcile

The recorded run hash differs from the current pipeline and notebook. The exact executed notebook snapshot is missing.

Recorded full-run code SHA-256
3bd6cb69a8bfeca0a63bf91c461152c45312a7e5c3b9ff55308d7090009b997e

Current scripts/course_pipeline.py SHA-256
a01bdafcf50131f31f1cc5bca963e66cfa6f9a6ac4240bbada93a6ec33fcdf95

Next evidence: archive the executed notebook, audit sequence similarity reproducibly, then evaluate locked choices on an external cohort.

Source-domain generalization, label quality and only 61 positive test cases limit the interpretation.

# Conclusions and next experiments

## Observed lessons under this dataset and protocol

| Question | Observed lesson |
| --- | --- |
| Composition sufficient? | Insufficient in this dataset |
| Ordered supervised patterns? | One-hot MCNN is a strong efficiency reference |
| Pretrained features? | Useful signal, but pooled classification is limited |
| Local extraction? | CNN improves over pooled 8M features here |
| Multiple scales? | Higher observed AP, with added head capacity |

## Next: similarity-aware validation and an external cohort with locked choices

35M MCNN has the highest observed test AP. Close model differences remain unresolved.

# Complete CV configurations (1/2)

## Appendix A1: all 22 settings, including unselected configurations

| Model | Setting | Mean AP | SD AP | MCC at 0.5 | Params |
|---|---|---|---|---|---|
| Composition + LR | C=0.1 | 0.1286 | 0.0631 | 0.1507 | 21 |
| Composition + LR | C=1 | 0.1287 | 0.0635 | 0.1505 | 21 |
| Composition + LR | C=10 | 0.1287 | 0.0635 | 0.1505 | 21 |
| One-hot + MCNN | k=2,3,4,5,8; F=64; weighted | **0.4943** | 0.1032 | 0.4182 | 28,801 |
| ESM 8M + pooled LR | C=0.1 | 0.3976 | 0.0572 | 0.3191 | 321 |
| ESM 8M + pooled LR | C=1 | 0.3264 | 0.0431 | 0.3173 | 321 |
| ESM 8M + pooled LR | C=10 | 0.2895 | 0.0397 | 0.2853 | 321 |
| ESM 8M + CNN | k=2; F=64; weighted | 0.4782 | 0.0607 | 0.3439 | 41,089 |
| ESM 8M + CNN | k=3; F=64; weighted | 0.4868 | 0.0600 | 0.3603 | 61,569 |
| ESM 8M + CNN | k=4; F=64; weighted | 0.4730 | 0.0692 | 0.3668 | 82,049 |
| ESM 8M + CNN | k=5; F=64; weighted | 0.4765 | 0.0427 | 0.3560 | 102,529 |

Mean and sample SD over saved outer-fold/seed rows. Selection reuses CV, so these are optimistic after tuning.

# Complete CV configurations (2/2)

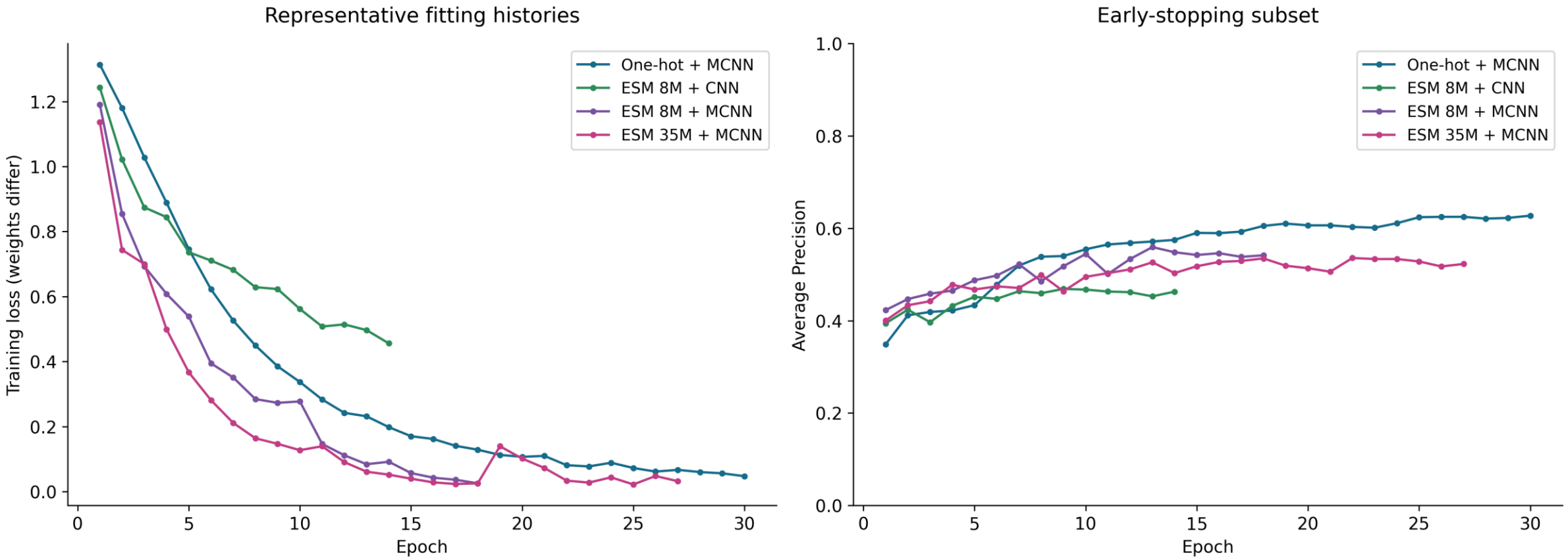
## Appendix A2: all 22 settings, including unselected configurations

| Model | Setting | Mean AP | SD AP | MCC at 0.5 | Params |
|---|---|---|---|---|---|
| ESM 8M + CNN | k=8; F=64; weighted | 0.4740 | 0.0413 | 0.3604 | 163,969 |
| ESM 8M + MCNN | k=2,3,4,5,8; F=32; weighted | 0.4922 | 0.0651 | 0.3718 | 225,601 |
| ESM 8M + MCNN | k=2,3,4,5,8; F=64; weighted | 0.5013 | 0.0551 | 0.4225 | 451,201 |
| ESM 8M + MCNN | k=2,3,4,5,8; F=128; weighted | 0.4985 | 0.0659 | 0.4280 | 902,401 |
| ESM 8M + MCNN | k=2,3,4,5,8; F=256; weighted | 0.5142 | 0.0547 | 0.4594 | 1,804,801 |
| ESM 8M + MCNN | k=2,3,4,5,8; F=64; unweighted | **0.4713** | 0.0384 | 0.4398 | 451,201 |
| ESM 35M + pooled LR | C=0.1 | 0.4449 | 0.0581 | 0.3712 | 481 |
| ESM 35M + pooled LR | C=1 | 0.3812 | 0.0319 | 0.3714 | 481 |
| ESM 35M + pooled LR | C=10 | 0.3193 | 0.0267 | 0.3631 | 481 |
| ESM 35M + CNN | k=3; F=64; weighted | 0.4913 | 0.0744 | 0.3769 | 92,289 |
| ESM 35M + MCNN | k=2,3,4,5,8; F=256; weighted | **0.5216** | 0.0655 | 0.4645 | 2,705,921 |

Mean and sample SD over saved outer-fold/seed rows. Selection reuses CV, so these are optimistic after tuning.

# Complete learning histories

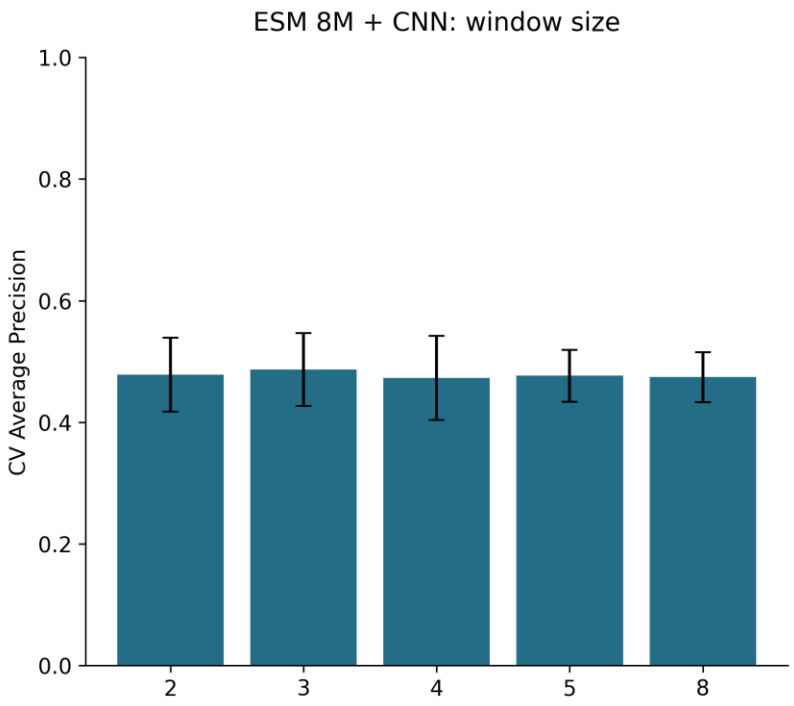
## Appendix A3: selected settings, first fold and first seed

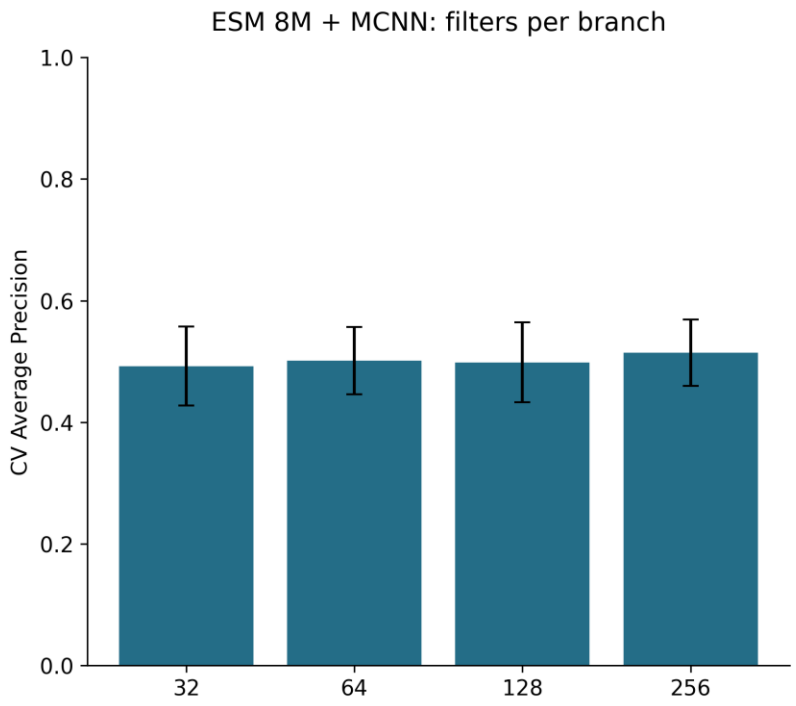


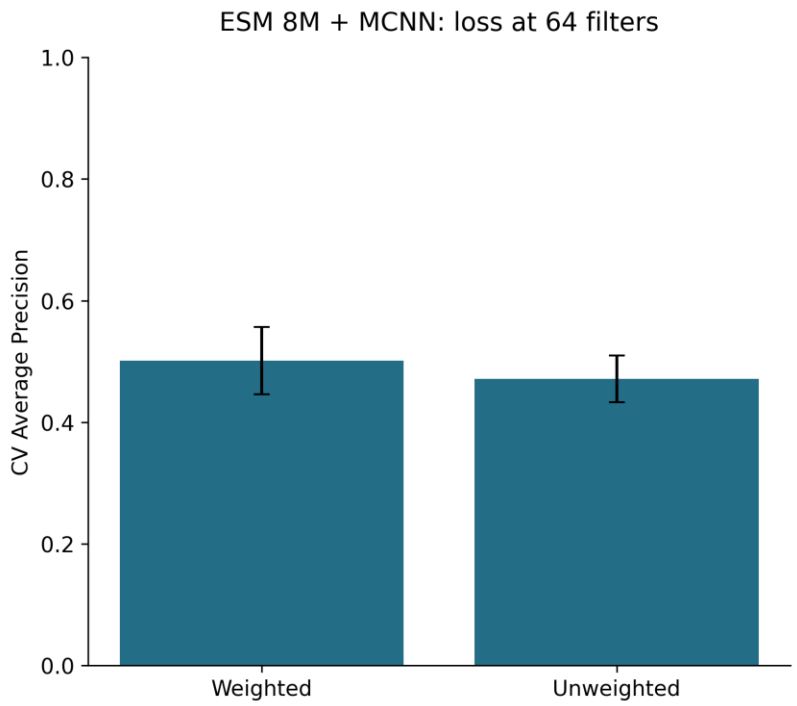

Fold 0, seed 535. Raw weighted losses have different scales. Early-stop AP supports checkpoint selection.

# Complete controlled ablations

## Appendix A4: one-factor 8M comparisons




Error bars show descriptive fold/seed SD. Window, filter and loss contrasts retain their fixed conditions.

# Additional representative errors

## Appendix A5: two FP and two FN examples for selected multi-window models

| Model | Record | Length | Label | Score | Threshold | Error |
|---|---|---|---|---|---|---|
| One-hot + MCNN | neo_sequence56 | 11 | 1 | **0.000960** | 0.512465 | FN |
| One-hot + MCNN | neo_sequence664 | 15 | 1 | 0.003104 | 0.512465 | FN |
| One-hot + MCNN | sgh_sequence28676 | 30 | 0 | 0.964475 | 0.512465 | FP |
| One-hot + MCNN | sgh_sequence19933 | 16 | 0 | 0.962445 | 0.512465 | FP |
| ESM 35M + MCNN | neo_sequence56 | 11 | 1 | **0.000162** | 0.632288 | FN |
| ESM 35M + MCNN | neo_sequence664 | 15 | 1 | 0.000451 | 0.632288 | FN |
| ESM 35M + MCNN | sgh_sequence7831 | 35 | 0 | **0.999930** | 0.632288 | FP |
| ESM 35M + MCNN | sgh_sequence7861 | 26 | 0 | 0.999847 | 0.632288 | FP |

Deterministic confidence-based examples, not a mechanistic explanation. Full sequences and all model examples remain in the source CSV.

# References and image credits

## Appendix A6: sources for biological context, methods and empirical figures

Chuang et al. (2024), International Journal of Biological Macromolecules
https://doi.org/10.1016/j.ijbiomac.2024.136252
Dataset and original project: https://github.com/B1607/DeepNeoAG

ESM models and implementation
https://github.com/facebookresearch/esm

Slide 2 illustration: adapted from Nature (2023) via National Cancer Institute
doi:10.1038/s41586-023-06063-y, CC BY 4.0. Resized only.
https://creativecommons.org/licenses/by/4.0/

Empirical figures: complete supplied training run, results_full/
Named figure exports: docs/assets/full_results/
Operating Review template: retained source deck, cloned and adapted

No AI-generated illustration is presented. Conceptual architecture placeholders remain for later image insertion.